MILITARY SPECIFICATION

BODY ASSEMBLY, TANK PERISCOPE: 10513599

This amendment forms a part of Military Specification MIL-B-60135(MU), dated 22 January 1965.

* Paragraph 2.1, under STANDARDS add:

"MIL-STD-810 Environmental Test Methods"

Paragraph 3.6.1.1: Delete paragraph and substitute:

"3.6.1.1 Operating temperatures.- There shall be no visible flicker of the image tubes while thermally stabilized at temperatures of -40°F and +125°F, when energized from an 18 to 30 vdc power source. The assembly shall be given a minimum of 5 minutes warm-up time before checking for flicker."

* Paragraph 3.6.2: Delete and substitute:

"3.6.2 Rigidity.- The catches shall be capable of securing the body assembly rigidly against the locating surfaces of the head and mount which it is intended to be used. This requirement shall be met during and subsequent to a vibratory exposure along 3 mutually perpendicular axes. The vibration shall be a simple harmonic motion; the frequency varying either linearly or logarithmically from 5 to 55 to 5 cps in 5 - minute cycles for a total of 15 minutes along each of the axes. The amplitude of vibration shall be maintained constant at 1/64-inch (1/32-inch double amplitude)."

Paragraph 3.9: Delete paragraph and substitute:

"3.9 Image steadiness.- There shall be no image movement or visible flicker subsequent to a minimum of 5 minutes stabilization or tube warm-up time, when the assembly is energized from an 18 to 30 vdc power source or a 1.5 vdc dry battery."

* Paragraphs 3.10, 3.10.1 and 3.10.2: Delete and substitute:

"3.10 Collimation.- The requirements of 3.10.1 and 3.10.2 shall be met under the following conditions:

(a) The top surface of the body housing in a horizontal plane.

(b) A target, 18 to 20 yards above the top surface of the body which lies on a vertical line perpendicular to a line joining the centers of the two objective apertures and at its midpoint.

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3.10.1 Dipvergence.- The angle formed by the two emergent lines of sight shall not exceed 15 minutes of arc.

3.10.2 Divergence.- The two emerging lines of sight shall diverge within the limits of 5 and 30 minutes of arc."

Paragraph 3.11: Delete paragraph and substitute:

"3.11 Image tilt.- The image of a target line originally perpendicular to a plane, defined by the 2 inch long shoulder (generated by the machining of the pads) adjacent to Valve Assembly 8201751, shall emerge perpendicular to the same plane within 45 minutes of arc."

Paragraph 3.13: Delete "7.0 minutes" and substitute "4.0 minutes".

Add the following new paragraph:

"3.13.1 Objective focus.- The objective assemblies shall be focused for the best image of a target at the center of the field, 18 to 20 yards above the top surface of the body. Best image is defined when the periscope body fulfils the resolution requirement."

Add the following new paragraphs:

"4.1.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) body assemblies plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.1.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

* Paragraph 4.2.2, Table I: Characteristic 108, change "4.5.3.7" to "4.5.3.7.1."

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* Paragraph 4.2.2, Table I: Add new characteristic, requirement and test:

"109. Objective focus 3.13.1 4.5.3.7".

Paragraph 4.3.2: Delete "Three body assemblies" and substitute "Ten body assemblies".

Paragraph 4.3.2: Following Characteristic 306, add new requirement and test:

"307. Operating temperatures 3.6.1.1 4.5.2.1.1"

Paragraph 4.4.2.3.2: Delete paragraph and substitute:

"4.4.2.3.2 Sealing.- Special testing equipment to test the sealing of the body assembly shall conform to the requirements of Drawing F8565556, Pressure Tester-Two Station."

* Paragraph 4.4.2.4: Delete and substitute:

"4.4.2.4 Device to test optical performance requirements.- The body assembly shall be tested for the performance requirements when mounted and secured to a testing device or fixture the accuracies of which shall conform to the pertinent features and accuracies depicted on Drawing F10527436 and drawings pertaining thereto shall apply.

a. Pretest information.- The test fixture shall be solidly supported on a leveled surface. This fixture shall also provide a power supply of 24 Vdc to illuminate the body assembly under test.

b. Set-up.- Place a one minute or better level across datum "A" shown on Fixture Drawing F10527436, Sheet 1, and by means provided adjust until test fixture is level. Then secure the fixture.

c. Suspend plumb lines in front of the double telescopes depicted on Drawing F10527436, Sheet 1, until the vertical center lines of the reticles are parallel to the plumb lines. The viewing unity power telescopes shall contain an erecting system.

d. Target.- A fabricated target shall be a square approximately 12 inches by 12 inches with a central vertical and horizontal line that shall cover the full length and width of the target square. The target shall contain white dull lines against a black absorbing background. The width of all target lines, excluding the resolution pattern lines shall be 2 minutes ± 30 seconds wide to permit good visibility.

e. Illumination.- The target shall be illuminated with two light sources containing 100 watt bulbs on Corning Glass #2563 filters or equivalent. The light sources shall be placed out of view of the line of sight but capable of providing sufficient illumination of the target. The resolution chart shall be centered about the geometric axis of the target.

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f. A qualified leveled surface for mounting the target shall be provided and it shall be so constructed to permit lateral movement of the target and maintain plumb of target image. The zero vertical line shall be alined plumb prior to any test accuracies of the body assembly. In addition, the target shall contain a rectangular pattern with acceptance limitation markings positioned from the left to the right of the zero vertical and horizontal lines. The acceptance limitation markings for divergence shall read 5 minutes to 30 minutes. Limitation markings for divergence shall read 15 minutes below and above the zero horizontal target image line. The fabricated target shall be positioned 56 feet 10 inches ± 3 feet from the front machined surface of the 45° mirror support conforming to Drawing F10527436, Sheet 6, and positioned as shown on Sheet 1 of Drawing F10527436. Its center shall be 1 inch ± 1/4 inch above the datum surface "A" and lie in the vertical plane passing the center of the holding fixture within 1/4 inch.

g. Viewing telescopes.- The emergent rays of the body assembly shall be measured with aid of the viewing telescopes Number 91-75-2 as shown on Drawing F10527436, alined to the geometric optical axis of the body eyepiece assemblies. The viewing telescopes line of sight shall be parallel and lay in the same horizontal plane. The vertical reticle lines of the telescopes shall be alined plumb when in the test position.

h. Mount the body assembly to text fixture. While viewing through the left unity power telescope readjust target for centering of the target on the reticle in the unity power telescope. Remove the telescopes. With the unaided eye adjust target for best focus of resolution pattern."

Following paragraph 4.5.2.1, add new paragraph:

"4.5.2.1.1 Operating temperatures.- The body assembly shall be exposed to the temperature conditions specified in 3.6.1.1, utilizing testing equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. The body assembly shall be placed in the cold chamber and the temperature of the chamber reduced gradually to -40°F. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period and while at -40°F, the body assembly shall be checked for flicker. The test shall be accomplished using the power supply capable of supplying the specified source. Apply a nominal 24 Vdc to the 24 volt receptical and place the double-pole double-throw switch in the 24 Vdc (ON) position. View through the eyepiece assembly of the periscope body and observe that there is no visible flicker of the image tube after being energized for the time period specified in 3.6.1.1. Upon completion of this test, the body assembly temperature shall be gradually raised to room ambient temperature. The body assembly shall then be placed into an oven chamber and the temperature shall be raised gradually to +125°F. The body assembly shall remain at this temperature for a period of 5 hours and at the conclusion of this 5 hour period, and while at +125°F, the body assembly shall again be subjected to the test depicted above to determine compliance with 3.6.1.1. Upon completion of this test the temperature of the test chamber shall be gradually reduced to room ambient temperature (60°F to 90°F) at

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which time the body assembly shall be removed from the test chamber. Caution shall be exercised during this test to avoid subjecting the body assembly to thermal shock.

NOTE: The rate of temperature change in the test chamber shall not exceed 20°F per hour throughout the temperature cycling test of 4.5.2.1 and 4.5.2.1.1."

* Paragraph 4.5.2.2: Delete and substitute:

"4.5.2.2 Rigidity.- Standard vibrating equipment shall be utilized for this test. The vibrating machine shall be equipped with means to secure a holding fixture as shown in Figure 1. Attach and clamp the body assembly in the holding fixture by means of the catches (see Figure 1). The test shall be performed in accordance with Method 514, Procedure 1X, Part 1 of MIL-STD-810 except that the conditions in 3.6.2 and as specified herein shall apply. The body assembly under test shall be vibrated along the three mutually perpendicular axes while the body assembly is maintained immobilized in the holding fixture. The amplitude of the applied vibration shall be monitored at the extreme bottom portion of the body assembly as positioned in the fixture. During vibration, check rigidity of the body assembly in the holding fixture using a stroboscope to produce the optical effect of stopping the vibration motion. The body assembly shall remain secure against the holding fixture receiving surfaces to assure conformance with the requirements of 3.6.2. At the conclusion of this test, the body assembly shall be subjected to the examination specified in 4.5.2.2.1."

* Add the following new paragraph:

"4.5.2.2.1 Examination.- Subsequent to rigidity (vibration), the body assembly shall be examined for any evidence of physical failure such as; A visual examination shall be made for casting failure, glass breakage, missing or damaged parts. A tactile examination shall be made of moving parts for undue irregularity of movement, friction or looseness. A functional examination of the catches shall be made to determine that the body assembly is clamped securely in the holding fixture, when the catches reach their apex and are fully closed. An audible examination shall be made of the body assembly for loose or broken internal parts such as the optical components or image converter tube, etc. Body assemblies that have satisfactorily passed aforementioned examinations shall be tested for and meet all the other requirements of this specification."

Paragraph 4.5.2.3: Delete paragraph and substitute:

"4.5.2.3 Sealing.- The internal pressurizing of the body assembly to test sealing, shall be performed with special testing equipment conforming 4.4.2.3.2. Pressurize the body assembly with dry nitrogen gas, to the value specified in 3.8 in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. Retain the applied pressure in the body assembly for a period of 5 minutes ± 15 seconds. At the end of the 5 minute period there shall be no evidence of leakage, as indicated on the pressure gauge of the testing equipment, to determine compliance with 3.8. The body assembly shall then be subjected to the test in 4.5.2.4."

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* Following paragraph 4.5.2.3, add new paragraph:

"4.5.2.4 Interior atmosphere.- A dew point tester shall be used to test the dew point of the interior atmosphere of the body assembly. Upon satisfactory completion of the sealing test, the internal pressure of the body assembly shall be reduced to 1 psig. The body assembly shall then be flushed with dry nitrogen gas. The flow of gas into the body assembly shall be controlled by a regulator that will prevent the internal pressure in the body assembly from exceeding 1 psig during the flushing operation. The flushing shall continue until the nitrogen gas bled from the body assembly has a dew point at least as low as minus 25°F, as measured with the dew point tester. The body assembly shall be sealed. The internal atmosphere of the body assembly shall conform to the requirements of 3.14."

Paragraph 4.5.3.1.1: Delete paragraph and substitute the following paragraphs in sequence:

"4.5.3.1.1 Image steadiness (18-30 Vdc).- Examination of body assemblies for no visible flicker or image movement as specified in 3.9, shall be performed on a 100% basis. The tests shall be performed utilizing the projector collimator inherent to the holding device. During the conduct of tests for acceptance inspection of image steadiness, the power source depicted in 4.5.3.1.1 shall be used to energize 75% of the body assemblies submitted for inspection and the power source depicted in 4.5.3.1.2 shall be used to energize the remaining 25%. The test shall be conducted using the power supply capable of supplying the specified source. Apply a nominal 24 Vdc to the 24 volt receptical and place the double-pole double-throw switch in the 24 Vdc (ON) position. View through the eyepiece assembly of the periscope body and observe that there is no movement of the target image or visible flicker of the image tube after being energized for the time period specified in 3.9.

4.5.3.1.2 Image steadiness (1.5 Vdc).- The test shall be performed concurrent with 4.5.3.1.1. Insert a 1.5 Vdc dry cell (C type) battery into the body assembly to energize the infrared tube. Place the double-pole double-throw switch to 1.5 volt (ON) position. Allow the time specified in 3.9 for stabilization of the target image. There shall be no visible flicker or movement of the target image to comply with 3.9."

* Paragraph 4.5.3.3: Delete and substitute:

"4.5.3.3 Dipvergence.- With the body assembly secured in the test fixture, sight through the right side and observe that the horizontal line of the viewing telescope is within the tolerance limits of the target image, and the requirements of 3.10.1."

* Paragraph 4.5.3.4: Delete and substitute:

"4.5.3.4 Divergence.- Observe through the right viewing telescope. The target center line shall be to the left of the vertical reticle line of the viewing telescope within the tolerance limits specified in 3.10.2 as measured on the target."

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* Paragraph 4.5.3.5: Delete and substitute:

"4.5.3.5 Image tilt.- Conduct the test by positioning the target to establish coincidence of the centerline (bottom) of the target image with respect to the vertical reticle line of the left viewing telescope. The opposite extremity of the telescope reticle shall fall within the limitation markings of the target image to determine conformance with the requirements specified in 3.11. Repeat the above procedure for the right side of the body assembly.

* Paragraph 4.5.3.7: Delete and substitute the following paragraphs:

"4.5.3.7 Objective focus.- This test shall be performed utilizing the resolution test target (4.4.2.4f) located the required distance from the body assembly. The objective focus will be at its best when the body assembly fulfills the resolution requirements of 3.13.1.

4.5.3.7.1 Resolution.- Perform this test by observing through each eyepiece assembly of the body assembly under test with the unaided eye. If there is any difficulty in observing the resolution chart an observation telescope with a magnification of not more than 3 power (3X) shall be utilized. The focus of this observation telescope shall be established at one best focus setting for viewing of the detectable line structure and proper line count of the resolution target. Resolution of each optical system shall meet the requirements specified in 3.13, under either viewing system described above. At the conclusion of resolution test insert the two unity power viewing telescopes to test fixture as shown on Drawing F10527436, Sheet 1. If necessary readjust the target while viewing through the unity power telescopes for centering the target on the left side of the reticle in the viewing telescope."

Paragraph 6.2: Add: "(e) Requirements for first article testing."

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